

Apple Sugar Contents Prediction via NIR-Style Image Generation: An Approach Based on an Extended CUT Model

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Abstract

This study presents a deep learning approach for the non-destructive prediction of apple sugar content using only RGB images. We enhance the Contrastive Unpaired Translation (CUT) model by integrating a physics-informed, domain-specific loss, which enables the network to better model the reflectance and brightness characteristics of near-infrared (NIR) images during the RGB-to-NIR translation process. The generated NIR-style images are applied to the AiHub apple dataset, which contains RGB images labeled with sugar content. These synthetic NIR images are then utilized to train an EfficientNet-based classifier for predicting sugar content. Experimental results indicate that our method achieves a significant improvement in prediction accuracy compared to models trained solely on RGB images, underscoring the effectiveness of the proposed domain-guided image translation strategy. Our findings demonstrate that it is possible to leverage NIR-like spectral information for fruit quality assessment without the need for specialized NIR sensors. By learning NIR spectral characteristics from RGB inputs, our approach facilitates more accurate and cost-effective classification of apple sugar content. This technique offers a practical solution for real-time, non-destructive fruit quality evaluation in agricultural settings where access to specialized sensors may be limited.

Keywords: Apple sugar contents prediction; RGB-to-NIR translation; Contrastive Unpaired Translation (CUT);

Domain-specific loss

1. Introduction

Fruit quality assessment is essential in modern agriculture, as it directly affects market value, consumer acceptance, and post-harvest management. Among various indicators, sugar content, typically expressed in Brix values, is one of the primary determinants of sweetness and overall quality. Conventional Brix measurement techniques can be categorized into two approaches: destructive and non-destructive. Destructive methods, such as refractometer measurements after juice extraction, provide reliable results but permanently damage the fruit, limiting their suitability for large-scale applications. In contrast, non-destructive techniques based on near-infrared (NIR) spectroscopy provide accurate measurements without harming the fruit. However, the high cost of NIR instruments and the requirement for specialized expertise restrict their widespread adoption in agricultural practice [1].

To address these limitations, recent studies have investigated the integration of computer vision and deep learning. In particular, unpaired image-to-image translation frameworks such as CycleGAN [2] and contrastive unpaired translation (CUT) [3] have been applied to generate NIR-style representations from RGB inputs, offering a cost-effective alternative to specialized NIR sensors. However, existing models often fail to capture key physical attributes of the NIR domain, including reflectance behaviour and brightness variation. Our work employs an imaging setup analogous to reflectance-based non-destructive sugar meters [1], ensuring that synthetic NIR images generated from RGB data remain consistent with established measurement principles.

Building on this foundation, we extend the CUT framework by introducing physics-informed, domain-specific loss functions, namely histogram matching, reflectance smoothness, and frequency-based loss, to capture the physical properties of NIR images more effectively. The proposed extended CUT framework is evaluated on both paired and unpaired datasets: a small RGB–NIR paired dataset of 60 apples for preliminary validation and the large-scale AiHub [4] RGB apple dataset for large-scale classification experiments. The AiHub dataset originally comprised 535,691 RGB apple images across four cultivars (Fuji, Hongro, Arisoo, and Sinano Gold) under multiple acquisition conditions. For this study, we used only Fuji samples captured in controlled indoor environments, excluding orchard scenes, to ensure consistency and measurement reliability. Synthetic NIR images produced by the extended CUT are used to train an EfficientNet-based classifier [5] for sugar content prediction.

Experimental results demonstrate that incorporating domain-specific loss functions substantially improves performance compared to the standard CUT model, increasing classification accuracy from 54% to 80% and yielding more balanced F1-scores across sugar content classes. These results confirm that synthetic NIR images derived from RGB inputs can enable accurate, non-destructive prediction of sugar content.

The remainder of this paper is organized as follows: Section 2 reviews related work on sugar content prediction and image translation; Section 3 describes datasets, preprocessing steps, and model design; Section 4 presents experimental results and analysis; and Section 5 concludes with implications and future research directions.

2. Related Work

2.1. Methods for sugar content prediction

Traditional approaches to measuring fruit sugar content can be categorized into two types: destructive and non-destructive. Destructive methods, such as refractometer measurements after juice extraction, provide accurate results but irreversibly damage the fruit, making them unsuitable for large-scale use. In contrast, non-destructive methods based on near-infrared (NIR) spectroscopy yield reliable and precise measurements. Still, they require costly equipment and skilled operation, which limits their accessibility in agricultural applications. These challenges underscore the need for scalable, cost-effective alternatives that preserve predictive accuracy without destructive sampling.

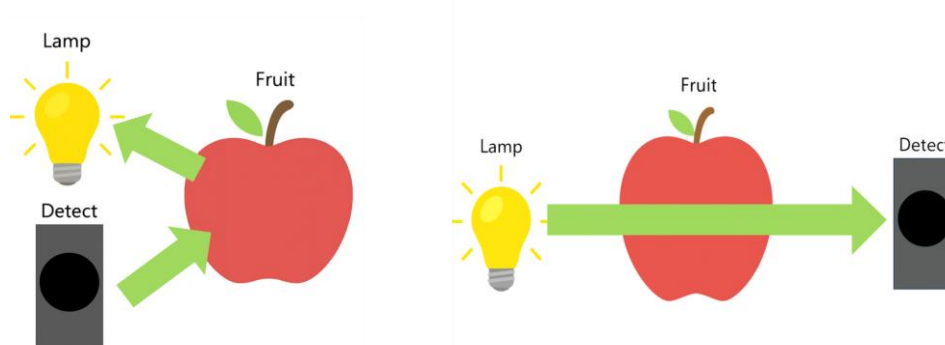


Figure 1. Schematic illustration of non-destructive sugar content measurement using NIR spectroscopy: (a) reflectance mode, and (b) transmittance mode [1].

As shown in Figure 1, NIR spectroscopy can be applied in two modes. In the reflectance mode (Figure 1a), light directed onto the fruit surface is reflected and captured by a detector, providing information about surface and near-surface tissues. In the transmittance mode (Figure 1b), light penetrates the fruit and is measured on the opposite side, yielding information about internal tissue composition. These two modes represent the fundamental physical principles of NIR-based non-destructive measurement.

In this study, we employed an imaging setup analogous to the reflectance mode, where NIR illumination was projected onto the apple surface and the reflected signals were recorded with a camera. This setup follows the same optical principle as reflectance-based non-destructive sugar meters, ensuring consistency with established practices. Furthermore, it highlights that generating synthetic NIR-style images from RGB inputs can be viewed as emulating reflection and transmission patterns, thereby enabling non-destructive sugar content prediction without the need for specialized NIR sensors.

Traditional color-based methods have shown potential but remain limited. Analyses of apple peel features in RGB, HSV, and Lab spaces revealed that color cues partly correlate with sweetness and enable basic classification. However, their accuracy is highly affected by lighting conditions and surface reflectance [6-8]. To address these shortcomings, the DeepNIR dataset [9] introduced large-scale paired RGB–NIR images, which enabled the training of generative models for creating synthetic NIR images and improving fruit detection performance. Together, these studies indicate two complementary directions: (1) exploiting color cues from RGB images and (2) generating synthetic NIR representations. Both approaches aim to provide scalable, non-destructive, and cost-efficient solutions for fruit quality prediction, laying the foundation for learning-based translation frameworks such as CycleGAN [2] and CUT [3].

2.2 Deep learning for image-to-image translation

Unpaired image-to-image translation has become essential in domains where paired datasets are scarce, such as RGB –NIR fruit imaging. CycleGAN [2] pioneered this field by introducing the cycle-consistency loss, which enabled bidirectional mappings without paired supervision. Despite its widespread adoption, CycleGAN often produced overly smoothed outputs with limited structural detail and diversity. CUT [3] improved this approach by

incorporating the PatchNCE loss. This contrastive learning strategy preserves fine-grained structures and enhances training stability by encouraging feature correspondence between input and translated images under unpaired conditions.

Nevertheless, both CycleGAN and CUT remain insufficient for spectral domains such as NIR, where accurately modelling reflectance and brightness variations is critical for realistic translation. These limitations underscore the need for domain-specific methods that more faithfully capture the spectral properties of NIR data, motivating the extended framework proposed in this study.

3. Methods

3.1 Dataset and preprocessing

3.1.1. Dataset description

The AiHub apple dataset [4], originally comprising 535,691 RGB apple images collected across four cultivars (Fuji, Hongro, Arisoo, and Sinano Gold) under diverse acquisition conditions, served as the primary resource for both RGB-to-NIR translation and sugar content classification. For this study, only Fuji apple samples captured in controlled indoor environments were used, while outdoor orchard images were excluded to ensure measurement reliability. Sugar levels were categorized into three classes based on Brix values: Low (<12 Brix), Mid (12–14 Brix), and High (>14 Brix). The low class contained substantially fewer samples than the Mid and high groups, creating class imbalance. To address this, we applied a cleaning process that removed outliers and excluded low-quality images, thereby achieving a more balanced representation of the data.

Additionally, we collected a paired dataset of 60 apples, comprising RGB and NIR images, for preliminary experiments. Although limited in size, this dataset enabled qualitative validation by allowing direct comparison of synthetic NIR images generated from RGB inputs with real NIR images, thereby confirming the feasibility of RGB-to-NIR translation.

3.1.2 Preprocessing steps

Several preprocessing procedures were applied before training and evaluation. First, images containing orchard backgrounds, top-view captures, or angled shots (approximately 45°) were excluded to ensure consistent representation of the fruit. Each apple image was then cropped to retain only the central fruit region and reduce background noise. Second, all images were resized to 256×256 pixels and normalized to stabilize GAN training. Third, the AiHub dataset was divided into training, validation, and test subsets using a fruit-level split, thereby preventing potential dataset leakage. Finally, labels were aligned with the Brix-based categories defined in Section 3.1.1, supporting classification and performance evaluation.

Through these steps, the AiHub dataset provided a scalable and standardized foundation for model training, while the paired RGB–NIR dataset enabled preliminary validation of NIR-style image generation..

3.2 Model architecture and loss functions

3.2.1 Baseline framework: CUT

The baseline framework was the CUT model [3], which extends the standard GAN structure composed of a generator G and a discriminator D . Unlike CycleGAN [2], which enforces global bidirectional mappings via cycle consistency, CUT employs the PatchNCE loss to preserve local structural details through contrastive learning.

Given an input RGB image $x \in X$, the generator produces a synthetic NIR image $\hat{y} = G(x)$. The discriminator D is trained to distinguish real NIR images $y \in Y$ and generated images $\hat{y}$. The adversarial loss is defined as:

$$\mathcal{L}_{GAN}(G, D) = \mathbb{E}_{y \sim p_{data}}(y)[\log D(y)] + \mathbb{E}_{x \sim p_{data}}(x)[\log (1 - D(G(x)))]$$

To ensure structural consistency between the input and generated images, CUT employs the PatchNCE loss, which is derived from the InfoNCE objective:

$$\mathcal{L}_{PatchNCE} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(f_q, f_k^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_q, f_j^-)/\tau)} \right]$$

where f_q denotes the query feature, f_k^+ the positive feature (corresponding to the same spatial location in the input and translated image), and f_j^- the negative features. The parameter τ controls the temperature parameter. In practice, this objective is applied across multiple network layers and spatial patch locations, yielding the expanded definition used in CUT:

$$\mathcal{L}_{PatchNCE}(G, H, X) = \mathbb{E}_{x \sim X} \left[\sum_{l=1}^L \sum_{s=1}^S -\log \frac{\exp(f_{l,s} \cdot f_{l,s}^+ / \tau)}{\sum_{j=1}^N \exp(f_{l,s} \cdot f_{l,n}^- / \tau)} \right]$$

where $f_{l,s}$ and $f_{l,s}^+$ denote positive pairs from the same spatial location across input and generated features, and $f_{l,n}^-$ are negative samples drawn from different locations. This formulation encourages each patch in the synthetic NIR image to align closely with its corresponding patch in the RGB input, thereby preserving fine-grained structures.

However, while CUT effectively maintains structural fidelity, it does not explicitly model spectral properties such as reflectance and brightness. To overcome this limitation, we extend CUT with domain-specific loss functions.

3.2.2 Extended CUT framework: Domain-specific losses

Although CUT [3] excels at preserving structural details, it neglects the physical properties inherent to NIR images.

We therefore introduce three additional loss functions:

- Histogram matching loss, which aligns intensity distributions to reduce global brightness discrepancies.
- Reflectance smoothness loss, which penalizes abrupt pixel-level variations and promotes local uniformity.
- Frequency-based loss, which constrains the frequency-domain representation to preserve fine-grained spectral details.

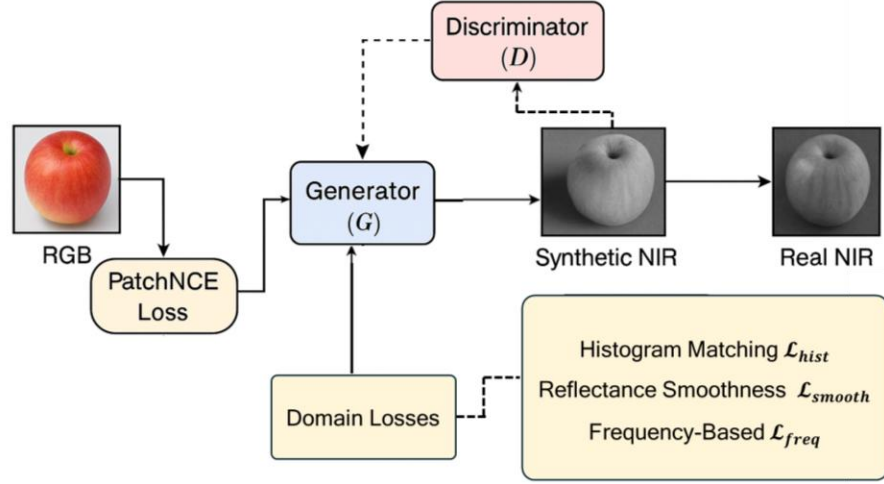


Figure 2. Overall framework of Extended CUT

The overall architecture of the proposed extended CUT framework is illustrated in Figure 2. The generator translates RGB inputs into synthetic NIR images, which the discriminator and the domain-specific loss modules then evaluate. Within the framework, the PatchNCE loss enforces structural consistency between the input and translated images, while the domain-specific losses ensure physical plausibility in terms of reflectance and brightness. These complementary objectives enable the generation of synthetic NIR images that closely approximate real NIR patterns, thereby enhancing the robustness of non-destructive fruit quality prediction in agricultural applications.

To clarify their roles, the proposed domain-specific losses are defined as follows.

- Histogram matching loss

$$\mathcal{L}_{hist} = \sum_{i=1}^B (H_y(i) - H_y(i))^2$$

where H_y and H_y denote the normalized histograms of the synthetic and real NIR images, respectively, with B bins. This loss aligns the global intensity distribution of synthetic NIR images with that of real NIR images, reducing discrepancies in brightness and contrast and yielding more realistic spectral representations.

- Reflectance smoothness loss

$$\mathcal{L}_{smooth} = \frac{1}{N} \sum_{x,y} (|I(x,y) - I(x+1,y)| + |I(x,y) - I(x,y+1)|)$$

where $I(x,y)$ is the pixel intensity at location (x,y) and N is the total number of pixels. This loss penalizes abrupt intensity changes between neighbouring pixels, promoting smooth reflectance transitions. Consequently, it preserves the local uniformity of real NIR images while suppressing artifacts such as noise and sharp edges.

- Frequency-based loss

$$\mathcal{L}_{freq} = \frac{1}{N} \sum_{u,v} |F_{\hat{y}}(u,v) - F_y(u,v)|$$

Where F denotes the two-dimensional Fourier transform of the image. This loss constrains the frequency-domain characteristics of synthetic NIR images to match those of real NIR images. By aligning spectral magnitudes, it preserves fine-grained details such as textures and structural boundaries that pixel-wise losses alone cannot capture.

3.2.3 Final objective function

These three domain-specific losses complement one another: histogram matching enforces global intensity consistency, smoothness loss constrains local reflectance, and frequency-based loss ensures spectral fidelity. Combined with adversarial and contrastive objectives from CUT, they form the complete optimization target:

$$\mathcal{L}_{total} = \mathcal{L}_{GAN} + \lambda_{nce} \mathcal{L}_{PatchNCE} + \lambda_1 \mathcal{L}_{hist} + \lambda_2 \mathcal{L}_{smooth} + \lambda_3 \mathcal{L}_{freq}$$

where $\lambda_{nce}, \lambda_1, \lambda_2, \lambda_3$ are hyperparameters that control the relative contributions of each loss term. This formulation integrates structural, photometric, and spectral constraints, allowing the generator to produce synthetic NIR images that closely approximate the characteristics of real NIR data.

3.3 Training procedure

3.3.1 Implementation details

All models were implemented in PyTorch [11] and trained using an NVIDIA GPU environment. The Adam optimizer was employed with a learning rate of 0.0002, $\beta_1 = 0.5$, and $\beta_2 = 0.999$, hyperparameter settings commonly adopted in GAN training to improve convergence stability. A batch size of four was used, and training was performed for 200 epochs. To further enhance stability, the learning rate was linearly decayed after the first 100 epochs. Data augmentation techniques, including horizontal flipping and random cropping, were applied to capture variations in fruit orientation and background, thereby improving model generalization.

The proposed framework was evaluated on two tasks.

- **Image translation quality:** RGB images were translated into synthetic NIR representations. The quality of the generated images was assessed both qualitatively, through visual inspection of structural fidelity and brightness patterns, and quantitatively using the structural similarity index (SSIM) [12] and the peak signal-to-noise ratio (PSNR).
- **Sugar content prediction:** Synthetic NIR images generated by the extended CUT framework were used to train an EfficientNet-based classifier [5]. The classifier predicted sugar content categories (low, mid, high), and performance was evaluated using accuracy, precision, recall, and F1-score. These metrics provided a comprehensive assessment of the framework's utility for predicting non-destructive fruit quality.

4. Experimental results and analysis

4.1. Evaluation metrics

To comprehensively evaluate the effectiveness of the proposed extended CUT framework, we employed complementary metrics that assessed both image translation quality and downstream classification performance.

4.1.1 Image translation quality

Three quantitative measures were used to evaluate the quality of RGB-to-NIR translation. The SSIM [12] quantifies structural fidelity between synthetic and ground-truth NIR images, with higher scores indicating better preservation of fine-grained details such as edges and contours. The PSNR evaluates pixel-level reconstruction accuracy, reflecting how closely the generated images match the intensity distribution of real NIR data. The fr chet inception distance (FID) [13] measures distributional similarity between synthetic and real NIR images in a deep feature space, thereby capturing perceptual realism beyond pixel-level metrics. Collectively, SSIM and PSNR capture local detail preservation, while FID complements them by quantifying global distributional alignment.

4.1.2 Sugar content classification

For the downstream classification task, four evaluation metrics were used: Accuracy, precision, recall, and F1-score. Accuracy measures the overall measure of correctness across all sugar content classes but may not adequately capture performance under class imbalance. Precision reflects resistance to false positives, whereas Recall emphasizes sensitivity to true positives. Their trade-off is fundamental in assessing fruit quality, where misclassification of low- or high-sugar apples can have practical consequences. The F1-score, defined as the harmonic mean of precision and recall, provides a balanced measure, especially useful for imbalanced datasets. By combining these metrics, we assessed not only the predictive capacity of the model but also its consistency across low, mid, and high sugar content classes.

Collectively, these metrics offer a comprehensive evaluation of the predictive utility of the proposed framework.

4.2 Qualitative image translation results

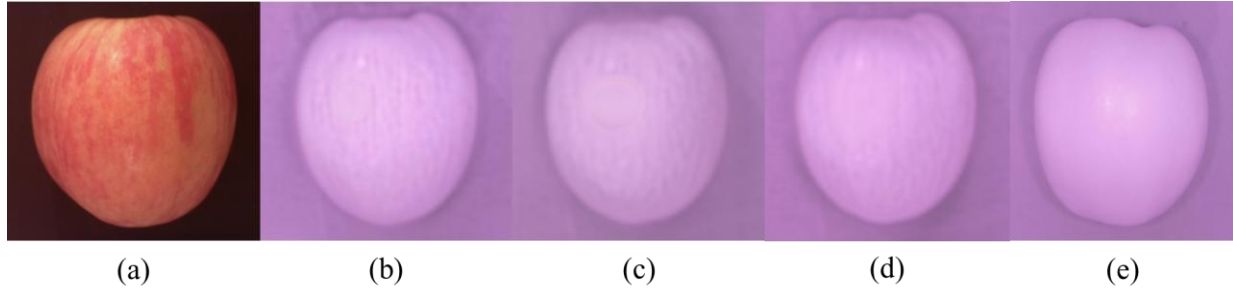


Figure 3. Representative examples of RGB-to-NIR translation for different models. From left to right: (a) input RGB image, (b) CycleGAN-generated NIR, (c) CUT-generated NIR, (d) Extended CUT-generated NIR (proposed method), and (e) real NIR reference.

As shown in Figure 3, CycleGAN produced overly smoothed images with blurred edges and reduced fine textures. This outcome is consistent with its lower SSIM and PSNR values, as well as its higher FID, indicating weaker structural fidelity and limited distributional alignment. CUT preserves structural features more effectively, as reflected in its higher SSIM. Still, it often fails to reproduce the brightness variations characteristic of the NIR domain, yielding only modest gains in PSNR. In contrast, the Extended CUT model, augmented with domain-specific loss functions, generates images with more realistic reflectance patterns and brightness distributions. These outputs exhibit a closer resemblance to real NIR references in both local structural details and global perceptual quality, which is supported by their higher SSIM and PSNR values and lower FID.

Overall, the qualitative and quantitative results indicate that the extended CUT framework generates synthetic NIR images with superior fidelity and realism. This enhanced representation establishes a stronger basis for downstream applications such as sugar content prediction, where both structural accuracy and domain-specific brightness modelling are essential. These findings further confirm that incorporating domain-specific losses effectively bridges the gap between RGB inputs and NIR spectral characteristics.

4.3 Sugar content classification results

To validate this in practice, we conducted preliminary experiments on a paired dataset of 60 apples, which included both RGB and NIR images. For each framework, RGB images were translated into synthetic NIR images, and an EfficientNet classifier was trained to assess classification accuracy quantitatively, in addition to qualitative validation against real NIR images.

Table 1. EfficientNet classification performance with real and synthetic NIR images

Model	Accuracy	Low F1	Mid F1	High F1
Real NIR	50%	0.00	0.33	0.50
CycleGAN	40%	0.50	0.29	0.44
CUT	54%	0.40	0.33	0.67
MUNIT	30%	0.00	0.39	0.29
DCLGAN	38%	0.57	0.00	0.17
DualCUT	38%	0.55	0.00	0.00

We compared several representative RGB-to-NIR translation frameworks. CycleGAN [2] introduced cycle consistency as an early solution; CUT [3] improved structural preservation through contrastive learning, and MUNIT [14] enabled multimodal generation to enhance output diversity. Table 1 summarizes the classification results obtained using synthetic NIR images generated by these models. Among them, CUT achieved the highest accuracy (54%) and relatively balanced F1-scores across the three sugar categories, whereas CycleGAN and MUNIT exhibited unstable performance and pronounced class imbalance. Real NIR images yielded only moderate accuracy, underscoring the inherent difficulty of predicting sugar content, even with authentic spectral data.

To extend this comparison, we also evaluated two recent CUT variants: DCLGAN [15] and DualCUT [16]. DCLGAN incorporates dual encoders to enhance semantic consistency, while DualCUT employs a dual contrastive learning strategy that aligns features bidirectionally between the source and target domains. In theory, these designs should improve the fidelity of synthetic NIR images and enhance classification outcomes. However, as shown in Table 1, neither DCLGAN nor DualCUT surpassed the original CUT. Both exhibited severe class imbalance, frequently misclassifying low or mid sugar categories and yielding unstable F1-scores. A plausible explanation is that the added architectural complexity over-constrained feature learning, limiting generalization

under imbalanced conditions. In fruit images with subtle reflectance variations, dual contrastive constraints may have reinforced local consistency at the expense of capturing the global structure necessary for accurate NIR translation.

These findings indicate that structural extensions alone cannot adequately model the essential properties of the NIR domain. Meaningful progress, however, requires domain-specific loss functions that explicitly encode NIR-specific characteristics, such as brightness distribution, tissue consistency, and reflectance patterns. This highlights the importance of extending CUT with physics-informed, domain-specific losses to capture NIR-style information for more faithful sugar content prediction.

4.4 Extended CUT with domain-specific loss

Building on the extended CUT framework described in Section 3.2.2, we incorporated three domain-specific losses, histogram matching, reflectance smoothness, and frequency-based loss, to better represent the physical properties of NIR images. These losses were combined with CUT’s GAN and PatchNCE objectives, forming the complete extended CUT model.

Because the additional losses operate at different scales, global intensity, local smoothness, and spectral magnitude, we introduced a weighting parameter λ to control their combined contribution to the total loss. Adjusting λ is critical to balance the trade-off: values that are too small λ under-regularize NIR priors, while overly large λ may oversmooth images or suppress discriminative features.

To systematically examine this effect, we empirically tested λ values in the range of $\{0.0, 0.1, 0.2, 0.5, 1.0\}$ based on preliminary trials, and evaluated performance using classification accuracy and F1-scores on the validation set. Although this procedure does not constitute a full grid search, it provided a structured basis to identify $\lambda=0.5$ as the most effective trade-off.

Table 2. Classification results of extended CUT with domain-specific losses under different λ values

λ_{domain}	Accuracy	Low F1	Mid F1	High F1
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0.0	30%	0.00	0.00	0.50
0.1	20%	0.00	0.33	0.25
0.2	40%	0.00	0.00	0.62
0.5	70%	0.80	0.80	0.40
1.0	50%	0.67	0.62	0.00

Table 2 summarizes classification performance across λ values. A clear concave trend emerges: very small λ (≤ 0.1) yields limited gains due to underweighted priors, while very large λ (> 1.0) reduces accuracy by over-constraining intensity and texture. At $\lambda = 0.5$, the model achieves the best trade-off, reaching 70% accuracy with balanced F1-scores across low, mid, and high classes. These findings confirm that domain-specific priors substantially improve downstream classification compared with the standard CUT.

Using the extended CUT model with $\lambda = 0.5$, we generated synthetic NIR images from the AiHub dataset. After preprocessing and manual filtering, the dataset contained 9,079 high-sugar, 6,087 mid-sugar, and 1,918 low-sugar samples. These images were then used to train and evaluate an EfficientNet classifier. Compared with the baseline CUT model, which achieved 54% accuracy (Table 1), the extended CUT delivered a notable improvement, achieving an overall accuracy of 80%.

Table 3. Class-wise performance of extended CUT with $\lambda = 0.5$

Class	Precision	Recall	F1-score
Low	0.85	0.53	0.66
Mid	0.78	0.91	0.84
High	0.83	0.76	0.80
Accuracy			80%

Table 3 summarizes the class-wise performance for $\lambda = 0.5$. The mid-class achieved the highest F1-score of 0.84, supported by a strong recall of 0.91, which reflects the benefit of having abundant training data. The High class also demonstrated robust performance, with an F1-score of 0.80, precision of 0.83, and recall of 0.76. In contrast, the low class achieved a high precision of 0.85 but a limited recall of 0.53, indicating that while predictions for the low category were mainly correct, many true low cases were not detected. This limitation directly stems from

dataset imbalance, as the low class contained only 1,918 samples compared with 9,079 and 6,087 in the other categories.

These findings demonstrate that incorporating domain-specific losses into CUT substantially enhances classification performance on large-scale datasets, increasing accuracy from 54% to 80%.

5. Conclusion and future work

This study introduced a deep learning framework for non-destructive prediction of apple sugar content using RGB images. By extending the CUT model with physics-informed, domain-specific losses, histogram matching, reflectance smoothness, and frequency-based constraints, we enabled the generation of synthetic NIR images that more accurately capture the physical properties of real NIR measurements. These synthetic images were then used to train an EfficientNet-based classifier, which achieved substantial improvements in classification accuracy.

The experimental results yielded three key insights. First, existing translation frameworks such as CycleGAN, CUT, and MUNIT exhibited limited effectiveness for sugar content prediction, with CUT providing the strongest baseline at 54% accuracy. Second, initial experiments with paired RGB–NIR data confirmed the feasibility of RGB-to-NIR translation but also underscored the difficulty of directly predicting sugar content even with authentic NIR inputs. Third, integrating domain-specific losses substantially enhanced performance: the extended CUT framework achieved 80% accuracy and more balanced F1-scores across sugar classes, underscoring the importance of explicitly modelling reflectance and brightness priors.

Despite these advances, several limitations remain. Class imbalance, particularly the underrepresentation of low-sugar samples, constrained recall performance and suggests the need for advanced balancing techniques or synthetic data augmentation. Additionally, while the proposed losses capture important physical priors, future studies could incorporate further constraints derived from optical models or leverage multimodal information (e.g.,

hyperspectral or thermal imaging) to improve fidelity. Finally, real-world deployment requires validation across diverse fruit varieties and environmental conditions to ensure robustness beyond the AiHub dataset.

In summary, this research demonstrates that physics-informed extensions to GAN-based translation frameworks can bridge the gap between RGB inputs and NIR spectral characteristics, providing a scalable and cost-effective alternative to traditional non-destructive sensors. Future research will aim to mitigate dataset imbalance, integrate richer domain knowledge, and expand generalization to broader agricultural applications.

References

1. So Minhyuk (2020) *A real-time non-destructive system implementation for the SSC measurement and the defected fruit detection algorithm based on M-CNN*. Dissertation, Cheongju University.
2. Zhu JY, Park T, Isola P, Efros AA (2017) *Unpaired image-to-image translation using cycle-consistent adversarial networks*. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp 2223–2232. <https://doi.org/10.1109/ICCV.2017.244>
3. Park T, Efros AA, Zhang R, Zhu JY (2020) Contrastive learning for unpaired image-to-image translation. In: *Computer Vision – ECCV 2020. Lecture Notes in Computer Science*, vol 12356. Springer, Cham, pp 319–345. https://doi.org/10.1007/978-3-030-58545-7_19
4. Jeonbuk Jangsu apple sugar content quality dataset. AI Hub (Korea), Apple Sugar Content Dataset. Available at: <https://www.aihub.or.kr/aidata/71593> (Accessed 3 September 2025)
5. Mingxing Tan, Quoc V. Le (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *Proceedings of the 36th International Conference on Machine Learning (ICML)*, PMLR 97: 6105–6114
6. Kim SJ (2020) *Association analysis of sugar contents based on color information of apple fruit images*. *Journal of KIIT* 18(4):41–49. <https://doi.org/10.14801/jkiit.2020.18.4.41>
7. Kim SJ (2022) *Analysis of apple colors and sugar contents using linear regression*. *JCCT* 8(1):201–207. <https://doi.org/10.17703/JCCT.2022.8.1.201>

8. Lee EJ, Min JI (2022) *Exploring influential color channels for CNN-based classification of apple quality grades. Journal of Digital Contents Society* 23(8):1477–1484. <https://doi.org/10.9728/dcs.2022.23.8.1477>
9. Zhang Y, Wang L, Li H, et al. (2021) DeepNIR: A large-scale dataset for near-infrared and RGB image translation in agriculture. *Computers and Electronics in Agriculture* 190:106433. <https://doi.org/10.1016/j.compag.2021.106433>
10. Sa I, Lim JY, Ahn HS, MacDonald B (2022) deepNIR: Datasets for generating synthetic NIR images and improved fruit detection system using deep learning techniques. *Sensors* 22(13):4721. <https://doi.org/10.3390/s22134721>
11. Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, Killeen T, Lin Z, Gimelshein N, Antiga L, Desmaison A, Köpf A, Yang E, DeVito Z, Raison M, Tejani A, Chilamkurthy S, Steiner B, Fang L, Bai J, Chintala S (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, Vancouver, Canada
12. Wang Z, Bovik AC, Sheikh HR, Simoncelli EP (2004). *Image quality assessment: From error visibility to structural similarity.* *IEEE Transactions on Image Processing* 13(4):600–612. <https://doi.org/10.1109/TIP.2003.819861>
13. Heusel M, Ramsauer H, Unterthiner T, Nessler B, Hochreiter S (2017). GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Advances in Neural Information Processing Systems (NeurIPS 2017)*, Long Beach, CA, USA, pp 6626–6637
14. Huang X, Liu M-Y, Belongie S, Kautz J (2018). *Multimodal Unsupervised Image-to-Image Translation.* In: *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 172–189. https://doi.org/10.1007/978-3-030-01219-9_11
15. Han J, Shoeiby M, Petersson L, Armin MA (2021). *Dual Contrastive Learning for Unsupervised Image-to-Image Translation.* In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 746–755. <https://doi.org/10.1109/CVPRW53098.2021.00086>
16. Yang B, Liu X, Li Y, Jin H, Qu Y (2024). *Multi-attention bidirectional contrastive learning method for unpaired image-to-image translation.* *PLOS ONE* 19(4):e0301580. <https://doi.org/10.1371/journal.pone.0301580>